



REPUBLIC OF YEMEN
SANA'A UNIVERSITY
FACULTY OF ENGINEERING
MECHATRONICS ENGINEERING DEPARTMENT



Exam

Done by:

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The first group

System(Parallel)

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Q4: For the full wave rectifier circuit shown in figure3 find the dc component and the coefficients of the for the harmonic signals for 8 terms.

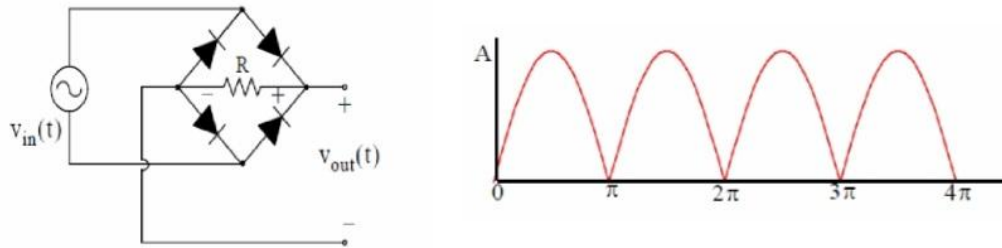


Figure 2 full wave rectifier and its output

Solutions:

#_The input is:-

```
>> % Define time points
t = linspace(0,4*pi,1000);
% Define the rectified waveform
function
f = abs(sin(t));
% Period is 4pi
T = 4*pi;
% Calculate DC component
dc = (1/T)*trapz(f);
% Calculate harmonic coefficients up
to 8th term
a = zeros(1,9);
for n = 1:8
    a(n) = (2/T)*trapz(f*cos(n*t/T));
end
% Print results
disp('DC component:')
disp(dc)
disp('Harmonic coefficients:')
disp(a)
```

#_The output is:-

DC component:
50.6099

Harmonic coefficients:
Columns 1 through 3

0.0466	0.1441	0.2037
--------	--------	--------

Columns 4 through 6

0.1706	0.0747	0.0042
--------	--------	--------

Columns 7 through 9

0.0264	0.1253	0
--------	--------	---

Q8: plot the frequency response of the system given by the transfer function:

$$H(\omega) = \frac{100(1 + j0.01\omega)}{(1 + j0.02\omega)(1 + j0.05\omega)}$$

Solutions:-

#_The input code:-

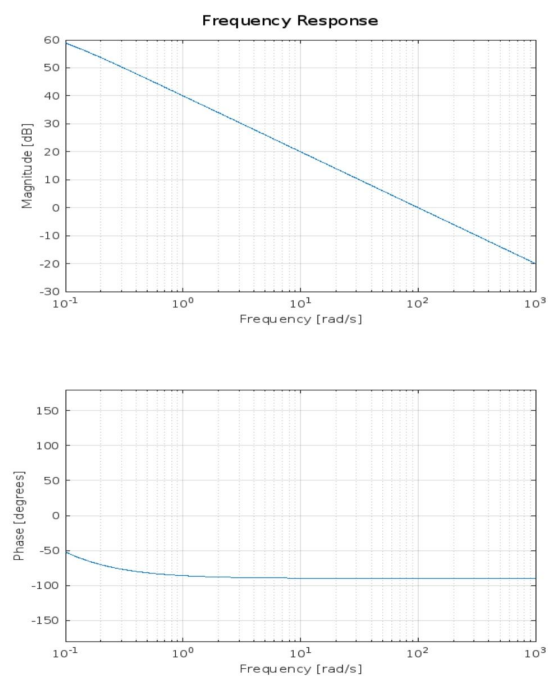
```
>> % Define the transfer function
numerator = [100 0.01];
denominator = conv([1 0.02], [1
0.05]);
% Create a frequency range
w = logspace(-1, 3, 1000);
% Compute the frequency response
h = freqs(numerator, denominator, w);
% Plot the magnitude response
subplot(2, 1, 1)
semilogx(w, 20*log10(abs(h)))
xlabel('Frequency [rad/s]')
ylabel('Magnitude [dB]')
grid on
% Plot the phase response
subplot(2, 1, 2)
semilogx(w, rad2deg(angle(h)))
xlabel('Frequency [rad/s]')
ylabel('Phase [degrees]')
grid on
% Adjust the plot limits for better
visibility
ylim([-180 180])
% Display the plot
sgtitle('Frequency Response')
```



>> Enter command here

##_The output is:-

Figure 1: % Define the transfer function numerator = [100 0.01]; denomina...



Q5: for the circuit shown in fig (4) the voltage is the input while the source current is the output of the system.

1. Find the transfer function of the system.
2. Compute the pole and zeros.
3. Express the transfer function in a partial fraction form.
4. Check the stability of the system (justify your answer).
5. Using Simulink show the step response and statistical measurements.

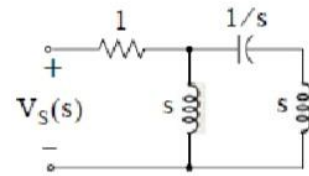


Figure 4

Hint: use pretty and simplify for clear results.

Solutions:-

#_The input is:-

##_The output is:-

Q6: Design a GUI that has the following features

1. On and off buttons.
2. The user can enter the transfer function.
3. The user can choose s or z domain and plot the s or z plane locating zeros and poles and check the stability of the system.
4. The user can simulate the step and frequency response of the system.

Solutions:-

#_The input is:-

```
function varargout = qoution_6(varargin)
% QOUTION_6 MATLAB code for qoution_6.fig
%   QOUTION_6, by itself, creates a new QOUTION_6 or raises the
%   existing
%   singleton*.
%
%   H = QOUTION_6 returns the handle to a new QOUTION_6 or the
%   handle to
%   the existing singleton*.
%
%   QOUTION_6('CALLBACK',hObject,eventData,handles,...) calls the
%   local
%   function named CALLBACK in QOUTION_6.M with the given input
%   arguments.
%
%   QOUTION_6('Property','Value',...) creates a new QOUTION_6 or
%   raises the
%   existing singleton*. Starting from the left, property value
%   pairs are
%   applied to the GUI before qoution_6_OpeningFcn gets called. An
%   unrecognized property name or invalid value makes property
%   application
%   stop. All inputs are passed to qoution_6_OpeningFcn via
%   varargin.
%
%   *See GUI Options on GUIDE's Tools menu. Choose "GUI allows only
%   one
%   instance to run (singleton)".
%
% See also: GUIDE, GUIDATA, GUIHANDLES

% Edit the above text to modify the response to help qoution_6

% Last Modified by GUIDE v2.5 20-Oct-2023 01:58:31

% Begin initialization code - DO NOT EDIT
gui_Singleton = 1;
gui_State = struct('gui_Name',       mfilename, ...
                  'gui_Singleton',   gui_Singleton, ...
                  'gui_OpeningFcn', @qoution_6_OpeningFcn, ...
                  'gui_OutputFcn',  @qoution_6_OutputFcn, ...
                  'gui_LayoutFcn',   [] , ...
                  'gui_Callback',    []);
if nargin && ischar(varargin{1})
    gui_State.gui_Callback = str2func(varargin{1});
end

if nargout
    [varargout{1:nargout}] = gui_mainfcn(gui_State, varargin{:});
else
    gui_mainfcn(gui_State, varargin{:});
end
% End initialization code - DO NOT EDIT

% --- Executes just before qoution_6 is made visible.
```

```

function qoution_6_OpeningFcn(hObject, eventdata, handles, varargin)
global transferferFunction;
global selectedDomain;
global domainValue;
transferferFunction='';
selectedDomain ='s';
domainValue =0;
% This function has no output args, see OutputFcn.
% hObject    handle to figure
% eventdata  reserved - to be defined in a future version of MATLAB
% handles     structure with handles and user data (see GUIDATA)
% varargin    command line arguments to qoution_6 (see VARARGIN)

% Choose default command line output for qoution_6
handles.output = hObject;

% Update handles structure
guidata(hObject, handles);

% UIWAIT makes qoution_6 wait for user response (see UIRESUME)
% uiwait(handles.figure1);

% --- Outputs from this function are returned to the command line.
function varargout = qoution_6_OutputFcn(hObject, eventdata, handles)
% varargout  cell array for returning output args (see VARARGOUT);
% hObject    handle to figure
% eventdata  reserved - to be defined in a future version of MATLAB
% handles     structure with handles and user data (see GUIDATA)

% Get default command line output from handles structure
varargout{1} = handles.output;

% --- Executes on button press in onbutton.
function onbutton_Callback(hObject, eventdata, handles)
% hObject    handle to onbutton (see GCBO)
% eventdata  reserved - to be defined in a future version of MATLAB
% handles     structure with handles and user data (see GUIDATA)

% --- Executes on button press in offbutton.
function offbutton_Callback(hObject, eventdata, handles)
% hObject    handle to offbutton (see GCBO)
% eventdata  reserved - to be defined in a future version of MATLAB
% handles     structure with handles and user data (see GUIDATA)

function edit1_Callback(hObject, eventdata, handles)
global transferferFunction;
transferferFunction=get(hObject, 'string');
% hObject    handle to edit1 (see GCBO)
% eventdata  reserved - to be defined in a future version of MATLAB

```

```

% handles      structure with handles and user data (see GUIDATA)

% Hints: get(hObject,'String') returns contents of edit1 as text
%        str2double(get(hObject,'String')) returns contents of edit1 as
a double

% --- Executes during object creation, after setting all properties.
function edit1_CreateFcn(hObject, eventdata, handles)
% hObject      handle to edit1 (see GCBO)
% eventdata    reserved - to be defined in a future version of MATLAB
% handles      empty - handles not created until after all CreateFcns
called

% Hint: edit controls usually have a white background on Windows.
%         See ISPC and COMPUTER.
if ispc && isequal(get(hObject,'BackgroundColor'),
get(0,'defaultUiControlBackgroundColor'))
    set(hObject,'BackgroundColor','white');
end

% --- Executes on button press in s_domainbutton.
function s_domainbutton_Callback(hObject, eventdata, handles)
global selectedDomain;
global domainValue;
    selectedDomain = 's';
    domainValue = 0;
% hObject      handle to s_domainbutton (see GCBO)
% eventdata    reserved - to be defined in a future version of MATLAB
% handles      structure with handles and user data (see GUIDATA)

% --- Executes on button press in z_domainbutton.
function z_domainbutton_Callback(hObject, eventdata, handles)
global selectedDomain;
global domainValue;
    selectedDomain = 'z';
    domainValue = 1;
% hObject      handle to z_domainbutton (see GCBO)
% eventdata    reserved - to be defined in a future version of MATLAB
% handles      structure with handles and user data (see GUIDATA)

% --- Executes on button press in plotbutton.
function plotbutton_Callback(hObject, eventdata, handles)

global selectedDomain;
global domainValue;
global transferFunction;

    if isempty(transferFunction)

        disp('Please enter ths transfer function.');
```

```

        return;
    end

```

```

    if strcmp(selectedDomain, 's')
        disp('Plotting the s_domain graph...');
    else
        disp('Plotting the z_domain graph...');
    end

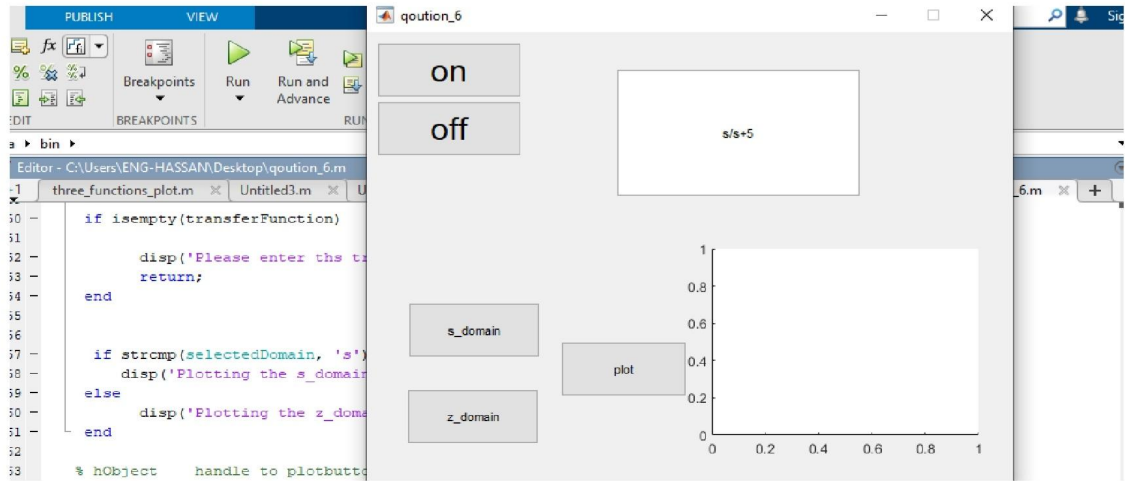
```

```

% hObject      handle to plotbutton (see GCBO)
% eventdata    reserved - to be defined in a future version of MATLAB
% handles      structure with handles and user data (see GUIDATA)

```

##_The output is:-



Q2. Using MATLAB to find $G(z)$ then plot the step and impulse response for:

1. $G(s) = \frac{s+3}{(s+0.5)(s-0.9)}$
2. $g(t) = n^2 u[n]$

Solutions:-

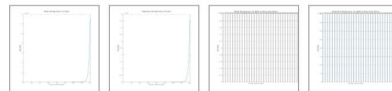
#_The input is:-

```
>> % Define the given transfer function
G(s)
numerator = [1 3];
denominator = [1 0.5 -0.9];
G_s = tf(numerator, denominator);
% Convert G(s) to G(z) using Tustin's
approximation with a sampling time of
Ts
Ts = 0.1; % Sampling time
G_z = c2d(G_s, Ts, 'tustin');
% Plot the step response of G(z)
figure;
step(G_z);
title('Step Response of G(z)');
xlabel('Time');
ylabel('Amplitude');
% Plot the impulse response of G(z)
figure;
impz(G_z);
title('Impulse Response of G(z)');
xlabel('Time');
ylabel('Amplitude');
% Define the given function g(t)
n = 2; % Constant value
t = 0:0.1:10; % Time vector
g_t = n^2 * (t >= 0); % Step function
% Convert g(t) to G(z) using ZOH
(Zero-Order Hold) method with a
sampling time of Ts
G_z_g_t = c2d(tf(g_t), Ts, 'zoh');
% Plot the step response of g(t) in
discrete-time
figure;
step(G_z_g_t);
title('Step Response of g(t) in
```

```

% Plot the step response of g(t) in
discrete-time
figure;
step(G_z_g_t);
title('Step Response of g(t) in
Discrete-time');
xlabel('Time');
ylabel('Amplitude');
% Plot the impulse response of g(t)
in discrete-time
figure;
impz(G_z_g_t);
title('Impulse Response of g(t) in
Discrete-time');
xlabel('Time');
ylabel('Amplitude');

```



>> Enter command here

##_The output is:-

Figure 2: % Define the given transfer function G(s) numerator = [1 3]; deno...

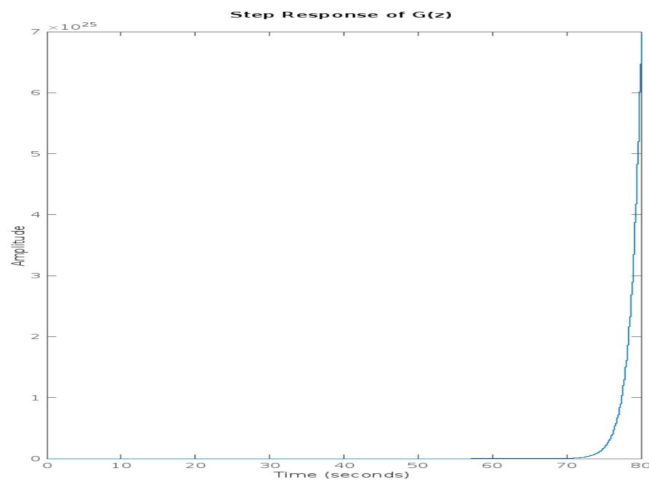


Figure 3: % Define the given transfer function G(s) numerator = [1 3]; deno...

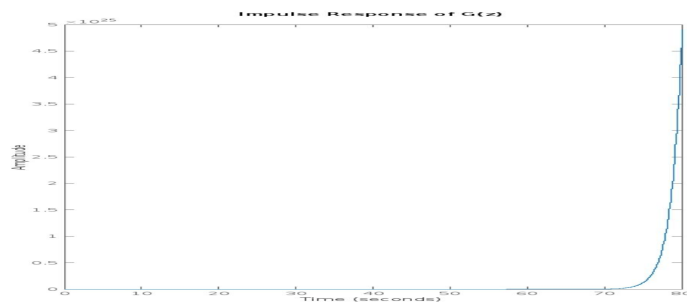


Figure 4: % Define the given transfer function G(s) numerator = [1 3]; deno...

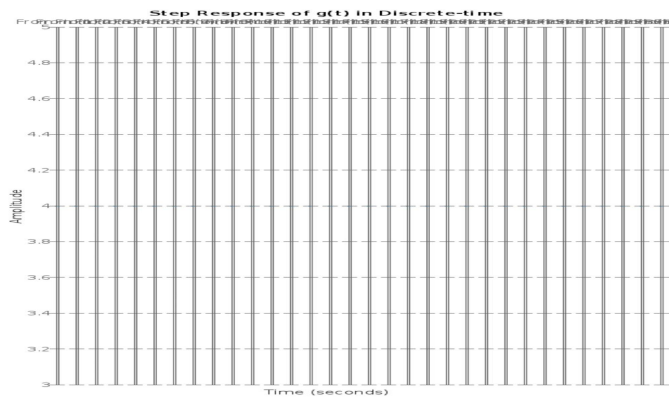
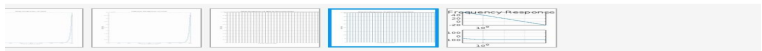
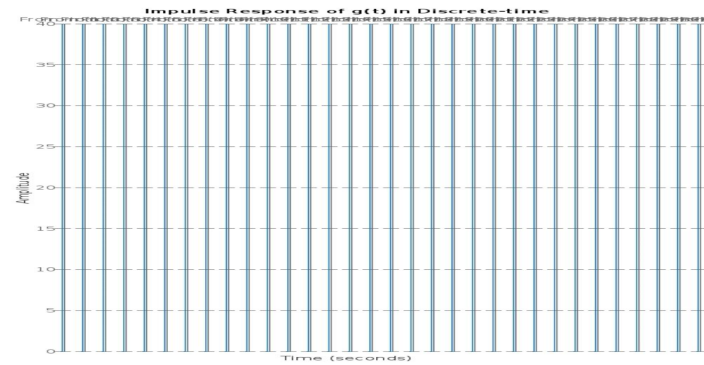
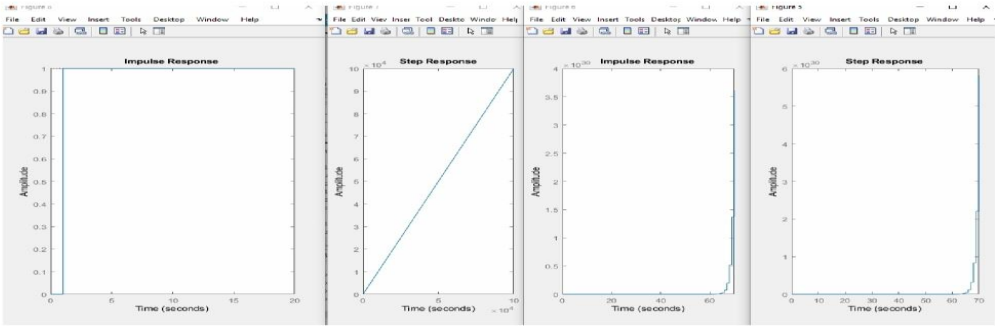
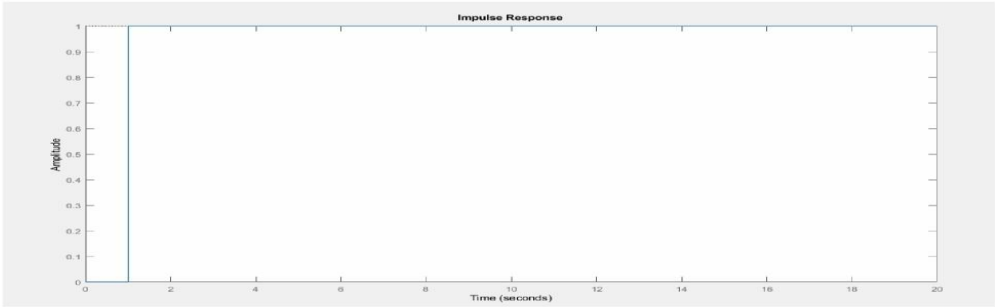


Figure 5: % Define the given transfer function G(s) numerator = [1 3]; deno...





Q1: Using Matlab plot the signal given by the function then save it in a pdf file.

$$y(t) = \begin{cases} 2t, & 0 < t < 1 \\ \frac{1}{2}(t^2 - 4t + 3), & 1 < t < 3 \\ 0, & \text{otherwise} \end{cases}$$

$$x[n] = 10(0.7)^n, n \geq 0$$

$$y[n] = 10(1.2)^n, n \geq 0$$

Solutions:-

#_The input is:-

##_The output is:-

Q3. A system is given by the block diagram

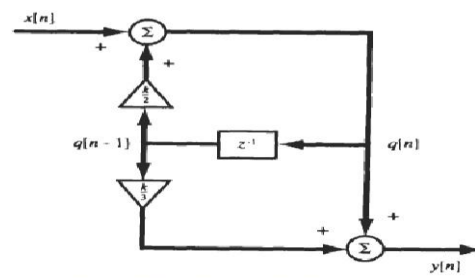


Figure 1 block diagram of a discrete system

If $k=-3$ and $k=3$ at which value of k the system is stable? plot the step and impulse response for both cases?

Solutions:-

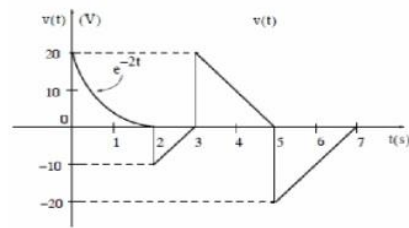
#_The input is:-

##_The output is:-

Q1: Using Matlab plot the signal then save it in a pdf file.

$$x[n] = 10(0.7)^n, n \geq 0$$

$$y[n] = 10(1.2)^n, n \geq 0$$



Solutions:-

#_The input is:-

##_The output is:-

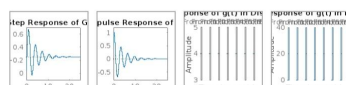
Q2. Using MATLAB to find $G(s)$ then plot the step and impulse response for:

1. $G(s) = \frac{z+1}{(z+0.5)(z+4)}$
2. $g(t) = n^2 u[n]$

Solutions:-

#_The input is:-

```
>> % Define the given transfer function
G(s)
numerator = [1 1];
denominator = [1 0.5 4];
G_s = tf(numerator, denominator);
% Convert G(s) to G(z) using the
bilinear transformation with a
sampling time of Ts
Ts = 0.1; % Sampling time
G_z = c2d(G_s, Ts, 'tustin');
% Plot the step response of G(z)
figure;
step(G_z);
title('Step Response of G(z)');
xlabel('Time');
ylabel('Amplitude');
% Plot the impulse response of G(z)
figure;
impz(G_z);
title('Impulse Response of G(z)');
xlabel('Time');
ylabel('Amplitude');
% Define the given function g(t)
n = 2; % Constant value
t = 0:0.1:10; % Time vector
g_t = n^2 * (t >= 0); % Step function
% Convert g(t) to G(z) using ZOH
(Zero-Order Hold) method with a
sampling time of Ts
G_z_g_t = c2d(tf(g_t), Ts, 'zoh');
% Plot the step response of g(t) in
discrete-time
figure;
step(G_z_g_t);
title('Step Response of g(t) in
Discrete-time');
% Plot the step response of g(t) in
discrete-time
figure;
step(G_z_g_t);
title('Step Response of g(t) in
Discrete-time');
xlabel('Time');
ylabel('Amplitude');
% Plot the impulse response of g(t)
in discrete-time
figure;
impz(G_z_g_t);
title('Impulse Response of g(t) in
Discrete-time');
xlabel('Time');
ylabel('Amplitude');
```



##_The output is:-

Figure 1: % Define the given transfer function G(s) numerator = [1 1]; deno... Figure 2: % Define the given transfer function G(s) numerator = [1 1]; deno...

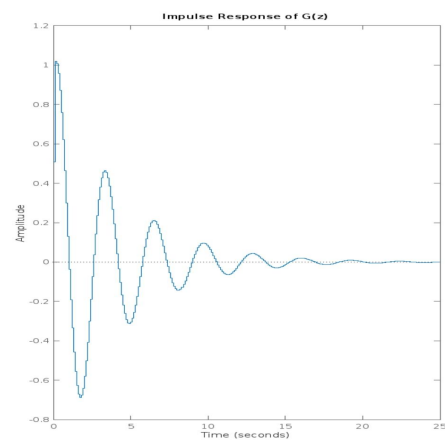
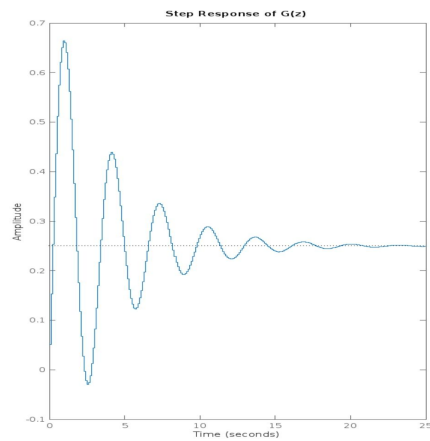


Figure 3: % Define the given transfer function G(s) numerator = [1 1]; deno...

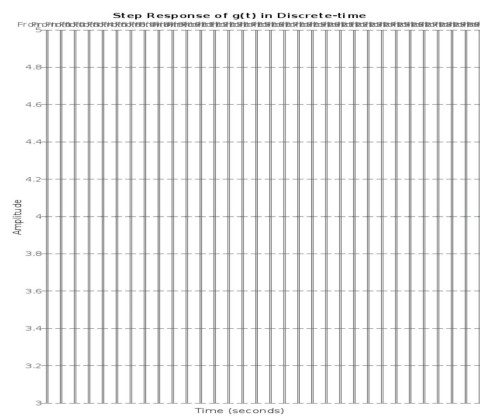
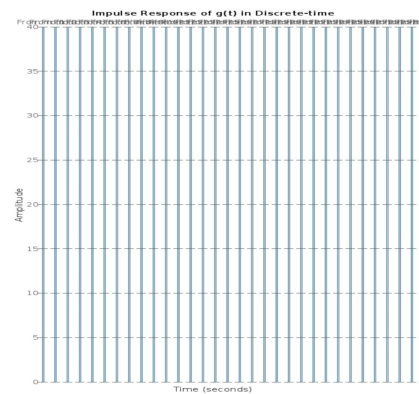


Figure 4: % Define the given transfer function G(s) numerator = [1 1]; deno...



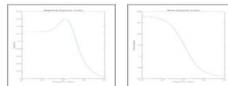
Q8: plot the frequency response of the system given by the transfer function

$$H(s) = \frac{s^2 + s + 1}{2s^3 + 10s^2 + 16s + 8} \quad s = j\omega$$

Solutions:-

#_The input is:-

```
>> % Define the transfer function H(s)
numerator = [1 1 -1];
denominator = [2 10 16 8];
H_s = tf(numerator, denominator);
% Define the frequency range
freq = logspace(-2, 2, 1000); %
Frequency range from 0.01 to 100 rad/
s
% Compute the frequency response
H_freq = freqresp(H_s, freq);
% Plot the magnitude response
figure;
semilogx(freq, abs(squeeze(H_freq)));
title('Magnitude Response of H(s)');
xlabel('Frequency (rad/s)');
ylabel('Magnitude');
% Plot the phase response
figure;
semilogx(freq, angle(squeeze(H_freq))
* 180 / pi);
title('Phase Response of H(s)');
xlabel('Frequency (rad/s)');
ylabel('Phase (degrees)');
```



>> Enter command here

##_The output is:-

